

Erythromycin interferes with the fluorometric determination of urinary catecholamines.

Interaction With Other Medicaments:

Erythromycin use in patients who are receiving high doses of theophylline may be associated with an increase of serum theophylline levels and potential theophylline toxicity. In case of theophylline toxicity and/or elevated serum theophylline levels, the dose of theophylline should be reduced while the patient is receiving concomitant erythromycin therapy. When oral erythromycin is given concurrently with theophylline, there is also a significant decrease in erythromycin serum concentrations. The decrease could result in subtherapeutic concentrations of erythromycin.

Concomitant administration of erythromycin and digoxin has been reported to result in elevated digoxin serum levels. There have been reports of increased anticoagulant effects when erythromycin and oral anticoagulants were used concomitantly. Increased anticoagulation effects due to interactions of erythromycin with various oral anticoagulants may be more pronounced in the elderly.

Concurrent use of erythromycin and ergotamine or dihydroergotamine has been associated in some patients with acute ergot toxicity characterised by severe peripheral vasospasm and dysesthesia.

Erythromycin has been reported to decrease the clearance of triazolam and midazolam and, thus, may increase the pharmacologic effect of these benzodiazepines.

Increases in serum concentrations of the following drugs metabolised by the cytochrome P450 system may occur when administered concurrently with erythromycin: acenocoumarol, alfentanil, astemizole, bromocriptine, carbamazepine, cyclosporin, digoxin, dihydroergotamine, disopyramide, ergotamine, hexobarbitone, midazolam, phenytoin, quinidine, rifabutin, tacrolimus, terfenadine, theophylline, triazolam, valproate, and warfarin.

Appropriate monitoring should be undertaken and dosage should be adjusted as necessary. Caution when taken with terfenadine or astemizole because it is likely to result in an enhanced risk of cardiotoxicity. The metabolism of terfenadine and astemizole is significantly altered when either is taken concomitantly with erythromycin.

Rare cases of serious cardiovascular events have been observed, including Torsades de pointes, other ventricular arrhythmias and cardiac arrest. Death has been reported with the terfenadine/erythromycin combination. Elevated cisapride levels have been reported in patients receiving erythromycin and cisapride concomitantly. This may result in QT prolongation and cardiac arrhythmias including ventricular tachycardia, ventricular fibrillation and Torsades de pointes. Similar effects have been observed with concomitant administration of pimozide and clarithromycin, another macrolide antibiotic.

Pregnancy and Lactation:

The safety of erythromycin for use during pregnancy has not been established. Erythromycin crosses the placental barrier. Erythromycin also appears in breast milk.

Side Effects:

Skin and Subcutaneous Disorders

Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS & acute generalised exanthematous pustulosis (AGEP).

The most frequent adverse effects of erythromycin preparations are gastrointestinal, e.g. abdominal cramping and discomfort, and are dose-related. Nausea, vomiting and diarrhoea occur infrequently with usual oral doses.

Onset of pseudomembranous colitis symptoms may occur during or after antibiotic treatment.

There have been isolated reports of reversible hearing loss occurring chiefly in patients with renal insufficiency and associated with doses of erythromycin usually greater than 4g per day

Allergic reactions ranging from urticaria and mild skin eruptions to anaphylaxis have occurred. Skin reactions ranging from mild eruptions to erythema multiforme, Stevens-Johnson syndrome and toxic epidermal necrolysis have rarely been reported.

There are no reports implicating erythromycin products with abnormal tooth development, and only rare reports of damage to the blood and kidneys. There have been isolated reports of transient central nervous system side effects including confusion, hallucinations, seizures and vertigo; however, a cause and effect relationship has not been established.

Rarely, erythromycin has been associated with the production of ventricular tachycardia and torsades de pointes, in individuals with prolonged QT intervals. There have been isolated reports of chest pain, dizziness and palpitations, however, a cause and effect relationship has not been established.

Symptoms of hepatitis, hepatic dysfunction and/or abnormal liver function test results may occur.

Symptoms & Treatment of Overdose:

The symptoms of overdosage include hearing loss, severe nausea, vomiting and diarrhoea. In case of overdosage, erythromycin should be discontinued. Overdosage should be handled with the prompt elimination of unabsorbed drug and all other appropriate measures.

Erythromycin is not removed by peritoneal dialysis or hemodialysis.

Storage Condition:

Store below 30°C. Protect from light.

Pack Size:

Blister pack of 1x10's, 10x10's, 50x10's and 100x10's.

Product Registration Number:

MAL05041850ACZ

Further information can be obtained from your doctor or pharmacist.

ROYCE®

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ROYCE®

Erymin-400 Tablet
(Erythromycin Ethyl Succinate-Antibiotic)

Presentation:

A pink coloured oblong shape film-coated tablet with a score on one side.

Content:

Each film-coated tablet contains:

Erythromycin Ethyl Succinate equivalent to Erythromycin 400mg.

Pharmacological Information:

Erythromycin exerts its antimicrobial action by binding to the 50S ribosomal sub-unit of susceptible microorganisms and suppresses protein synthesis. It is active against most strains of the following organisms:

Gram Positive bacteria: *Listeria monocytogenes*, *Corynebacterium diphtheria*, Staphylococci spp and Streptococci spp (including Enterococci).

Gram Negative bacteria: *Haemophilus influenzae*, *Neisseria meningitidis*, *Neisseria gonorrhoeae*, *Legionella pneumophila*, *Moraxella (Branhamella) catarrhalis*, *Bordetella pertussis* and Campylobacter spp.

Mycoplasma: *Mycoplasma pneumonia* and *Ureaplasma urealyticum*.

Other organisms: *Treponema pallidum*, Clostridia spp, Chlamydia spp, L-forms, the agents causing trachoma and *lymphogranuloma venereum*.

Orally administered erythromycin ethyl succinate tablet is readily absorbed from the small intestine. Comparable serum levels of erythromycin are achieved in the fasting and non fasting states. Peak blood levels normally occur within 1 hour of dosing of erythromycin ethyl succinate tablet.

Erythromycin ethyl succinate is less susceptible than erythromycin to the adverse effect of gastric acid. It is widely distributed throughout body tissues and diffuses readily into most body fluids. Only low concentrations are normally achieved in the spinal fluid, but passage of the drug across the blood-brain barrier increases in meningitis. In the presence of normal hepatic function, erythromycin is concentrated in the liver and excreted in the bile; the effect of hepatic dysfunction on excretion of erythromycin by the liver into the bile is not known. Less than 5% of the orally administered dose of erythromycin is excreted in active form in the urine. The elimination half life is approximately 2 hours. Doses may be administered 2, 3 or 4 times a day. Erythromycin crosses the placental barrier and is excreted in breast milk.

Indication:

For the prophylaxis and treatment of infections caused by erythromycin-sensitive organisms. Erythromycin is highly effective in the treatment of a great variety of clinical infections such as:

1. Upper respiratory tract infections of mild to moderate degree caused by *Streptococcus pyogenes*, *Streptococcus pneumoniae*, or *Haemophilus influenzae* (when used concomitantly with adequate doses of sulfonamides, since many strains of *H. influenzae* are not susceptible to the erythromycin concentrations ordinarily achieved).

2. Lower respiratory tract infections of mild to moderate severity caused by *Streptococcus pneumoniae* or *Streptococcus pyogenes*.

3. Listeriosis caused by *Listeria monocytogenes*.

4. Pertussis (whooping cough) caused by *Bordetella pertussis*. Erythromycin is effective in eliminating the organism from the nasopharynx of infected individuals rendering them non-infectious. Some clinical studies suggested that erythromycin may be helpful in the prophylaxis of pertussis individuals.

5. Respiratory tract infections due to *Mycoplasma pneumoniae*.

6. Skin and skin structure infections of mild to moderate severity caused by *Streptococcus pyogenes* or *Staphylococcus aureus* (resistant staphylococci may emerge during treatment).

7. Diphtheria: Infections due to *Corynebacterium diphtheriae*, as an adjunct to antitoxin, to prevent establishment of carriers and to eradicate the organism in carriers.

8. Erythrasma: In the treatment of infections due to *Corynebacterium minutissimum*.

9. Intestinal amebiasis caused by *Entamoeba histolytica* (oral erythromycin only). *Extraenteric amebiasis* requires treatment with other agents.

10. Acute pelvic inflammatory disease caused by *Neisseria gonorrhoeae*: As an alternative drug in treatment of acute pelvic inflammatory disease caused by *N. gonorrhoeae* in female patients with a history of sensitivity to penicillin. Patients should have a serologic test for syphilis before receiving erythromycin as treatment of gonorrhea and a follow-up serologic test for syphilis after 3 months.

11. Syphilis caused by *Treponema pallidum*: Erythromycin is an alternate choice of treatment for primary syphilis in patients allergic to the penicillins. In treatment of primary syphilis, spinal fluid examinations should be done before treatment and as part of follow-up after therapy.

12. Erythromycins are indicated for the treatment of the following infections caused by Chlamydia trachomatis: conjunctivitis of the newborn, pneumonia of infancy, and urogenital infections during pregnancy. When tetracyclines are contraindicated or not tolerated, erythromycin is indicated for the treatment of uncomplicated urethral, endocervical, or rectal infections in adults due to Chlamydia trachomatis.

13. When tetracyclines are contraindicated or not tolerated, erythromycin is indicated for the treatment of nongonococcal urethritis caused by Ureaplasma urealyticum.

14. Legionnaire's Disease caused by Legionella pneumophila. Although no controlled clinical efficacy studies have been conducted, in vitro and limited preliminary clinical data suggest that erythromycin may be effective in treating Legionnaires' Disease.

15. Prophylaxis: Prevention of initial attacks of Rheumatic Fever: Penicillin is considered by the American Heart Association to be the drug of choice in the prevention of initial attacks of rheumatic fever (treatment of Streptococcus pyogenes infections of the upper respiratory tract, e.g., tonsillitis or pharyngitis). Erythromycin is indicated for the treatment of penicillin-allergic patients. The therapeutic dose should be administered for 10 days.

Prevention of Recurrent Attacks of Rheumatic Fever: Penicillin or sulfonamides are considered by the American Heart Association to be the drugs of choice in the prevention of recurrent attacks of rheumatic fever. In patients who are allergic to penicillin or sulfonamides, oral erythromycin is recommended by the American Heart Association in the long-term prophylaxis of streptococcal pharyngitis (for the prevention of recurrent attacks of rheumatic fever).

Prevention of Bacterial Endocarditis: Although no controlled clinical efficacy trials have been conducted, oral erythromycin has been recommended by the American Heart Association for prevention of bacterial endocarditis. In penicillin-allergic patients with prosthetic cardiac valves, most congenital cardiac malformations, surgically constructed systemic pulmonary shunts, rheumatic or other acquired valvular dysfunction, idiopathic hypertrophic subaortic stenosis (IHSS), previous history of bacterial endocarditis or mitral valve prolapse with insufficiency when they undergo dental procedures or surgical procedures of the upper respiratory tract.

Dosage and Administration:

For oral use only.

Erythromycin ethyl succinate may be administered without regard to meals.

Children: Age, weight and severity of the infection are important factors in determining the proper dosage. In mild to moderate infections the usual dosage of erythromycin ethyl succinate for children is 30-50mg/kg/day in equally divided doses every 6 hours. For more severe infections this dosage may be doubled. If twice-a-day dosage is desired, ½ of the total daily dose may be given every 12 hours. Doses may also be given 3 times daily by administering ⅓ of the total daily dose every 8 hours.

The following dosage schedule is suggested for mild to moderate infections:

Body Weight (lbs)	Total Daily Dose
<10	30-50mg/kg/day 15-25mg/kg/q 12h
10-15	200mg
16-25	400mg
26-50	800mg
51-100	1200mg
>100	1600mg

Adults: 400mg erythromycin ethyl succinate every 6 hours is the usual dose. Dosage may be increased up to 4g/day according to the severity of the infection. If twice-a-day dosage is desired, ½ of the total daily dose may be given every 12 hours. Doses may also be given 3 times daily by administering ⅓ of the total daily dose every 8 hours.

For adults' dosage calculation, use a ratio of 400mg of erythromycin activity as the ethyl succinate to 250mg of erythromycin activity as the stearate, base or estolate.

In the treatment of streptococcal infections, a therapeutic dosage of erythromycin ethyl succinate should be administered for at least 10 days. In continuous prophylaxis against recurrences of streptococcal infections in persons with a history of rheumatic heart disease, the usual dosage is 400mg twice a day.

For prophylaxis against bacterial endocarditis in patients with congenital heart disease, or rheumatic or other acquired valvular heart disease when undergoing dental procedures or surgical procedures of the upper respiratory tract, give 800mg (10mg/kg for children) orally 2 hours prior to the procedure and then, 400mg (5mg/kg for children) 6 hours after the initial dose.

For the treatment of urethritis due to C. trachomatis or U. urealyticum: 800mg 3 times a day for 7 days.

For treatment of primary syphilis: Adults: 48-64g given in divided doses over a period of 10-15 days. For intestinal amebiasis: Adults: 400mg 4 times daily for 10 – 14 days. Children: 30 – 50 mg/kg/day in divided doses for 10 –14 days.

For use in pertussis: Although optimal dosage and duration have not been established, doses of erythromycin utilized in reported clinical studies were 40 –50 mg/kg/day, given in divided doses for 5 –14 days.

For treatment of Legionnaires' Disease: Although optimal doses have not been established, doses utilized in reported clinical data were those recommended above (1.6 to 4g daily in divided doses).

Contraindication:

Patients with known hypersensitivity to erythromycin. Erythromycin is contraindicated in patients taking astemizole, terfenadine, cisapride or pimozide. It is contraindicated in patients with severely impaired hepatic function (see Interactions With Other Medicaments).

Warning & Precautions:

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) (e.g. Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)), Erymin-400 Tablet should be discontinued immediately and appropriate treatment should be urgently initiated.

Erythromycin is principally excreted by the liver. Caution should be exercised in administering the antibiotic to patients with impaired hepatic function or concomitantly receiving potentially hepatotoxic agents. There have been reports of hepatic dysfunction, with or without jaundice, occurring in patients receiving oral erythromycin products. During prolonged or repeated therapy, there is a possibility of overgrowth of non-susceptible bacteria or fungi. If such infections occur, the drug should be discontinued and appropriate therapy instituted.

There have been reports suggesting erythromycin does not reach the foetus in adequate concentrations to prevent congenital syphilis. Infants born to women treated during pregnancy with oral erythromycin for early syphilis should be treated with an appropriate penicillin regimen.

Pseudomembranous colitis has been reported with nearly all antibacterial agents, including erythromycin, and may range in severity from mild to life threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of antibacterial agents. Treatment with antibacterial agents alters the normal flora of the colon and may permit overgrowth of clostridia. Studies indicate that a toxin produced by Clostridium difficile is a primary cause of 'antibiotic-associated colitis'.

After the diagnosis of pseudomembranous colitis has been established, therapeutic measures should be initiated. Mild cases of pseudomembranous colitis usually respond to discontinuation of the drug alone. In moderate to severe cases, consideration should be given to management with fluids and electrolytes, protein supplementation, and treatment with an antibacterial drug clinically effective against Clostridium difficile colitis.

Rhabdomyolysis with or without renal impairment has been reported in seriously ill patients receiving erythromycin concomitantly with lovastatin. Therefore, patients receiving concomitant lovastatin and erythromycin should be carefully monitored for creatine kinase (CK) and serum transaminase levels.

There have been reports that erythromycin may aggravate the weakness of patients with myasthenia gravis.

Prolonged or repeated use of erythromycin may result in an overgrowth of nonsusceptible bacteria or fungi. If superinfection occurs, erythromycin should be discontinued and appropriate therapy instituted.

When indicated, incision and drainage or other surgical procedures should be performed in conjunction with antibiotic therapy.